Web Technologies (C2111C3) – Unit 1 Notes

World Wide Web Architecture · ITM (SLS) Baroda University · Dept. of CSE · 4 contact hours

Unit Roadmap

Hour	Theme	Topics
1	Foundations	WWW concept, Internet vs WWW, URIs/URLs, W3C, history
2	Protocols & Clients	HTTP/1.1 & HTTP/2, headers, status codes, browser rendering, web servers
3	Infrastructure	Web 2.0 (AJAX/REST), Linux server setup (Apache/Nginx), virtual hosts, dynamic IP & DDNS
4	Web Design	UI/UX, visual hierarchy, responsive design, accessibility (WCAG), performance

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1. Introduction to the World Wide Web (WWW)

- The **WWW (the Web)** is a system of interconnected web pages and information accessed using the **Internet**.
- It was created to help people **share and find information easily** using **links (hyperlinks)** that connect pages.
- Collectively, all public websites accessible through the internet are called the **W3 pool**.

Key characteristics

- **Global interaction** – browse sites, watch videos, shop online, connect with people worldwide.
- **Hyperlink navigation** – users discover more information by moving through interlinked pages.
- **Rich formats** – text, pictures, audio and video.
- The Web connects over **63% of the world's population**.

2. History of the WWW

- Created by **Tim Berners-Lee in 1989** at **CERN** so researchers could collaborate and share information.
- The **World Wide Web Consortium (W3C)** was founded to develop **common standards and specifications** so the web grows openly.

Evolution of the Web

Version	Period	Key idea
Web 1.0	1990–2000	Static websites, mostly **one-way** information sharing (read-only)
Web 2.0	2000–2010	Interactive, social platforms; **user-generated content**; community connection
Web 3.0	2010–2020	**Semantic web** – machines understand data; personalised experiences
Web 4.0	2020–2030	Expected to be intelligent, decentralised, AI-powered, with web operating systems

(The slide's evolution graph also shows a "Web 2.5" – mobile computing, virtualization, cloud computing – as the "real evolution" from 2010 onwards.)

Quick facts

- The first website is still live: `http://info.cern.ch/`
- 1.5 billion+ websites exist.
- Google Chrome is used by 60%+ of web users.
- The Web operates in 150+ languages.

3. Concept of the WWW

Web browser

A software interface used to **access and display hyperlinked resources** on the WWW, translating code into accessible multimedia.

- **Resource rendering** – text, data, pictures, animation, video.
- **Multi-tasking utility** – searches, file transfers, mailing services.

- **Examples** – Google Chrome, Opera Mini, Internet Explorer.

Client–server model

- The web works on a **client–server architecture**.
- **Web browser (client)** → sends a *request* to the **web server**.
- **Web server** → *serves the web page* back to the client.

4. Internet vs WWW

	The Internet	The World Wide Web
Layer	Hardware / network layer	Application / data layer
What it is	Globally decentralised **network of networks** (autonomous systems) running on **TCP/IP**	An **information system running on top of the Internet**
Handles	Routers, fiber optics, packet switching, IP routing, TCP/UDP	Hypertext, **URIs**, **HTTP** to interlink documents
Inventor	–	Sir Tim Berners-Lee, CERN (1989)

> **Key takeaway:** The Internet is the **infrastructure (roads)**; the WWW is the **service (vehicles and traffic on the roads)**.
>
> Other services on the Internet besides the web: email, FTP, NNTP (Usenet) etc.

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5. URL Anatomy

Example: `https://www.itmbu.ac.in`

Part	Value	Meaning
1. Scheme / protocol	`https://`	Protocol used to transfer data. The **s** = **Secure** (SSL/TLS encryption, protects from interception/tampering)

Part	Value	Meaning
2. Subdomain	`www`	Subdivision of the main domain; traditionally the primary public web server. Optional today but still a convention
3. Second-level domain (SLD)	`itmbu`	Organisation identifier (Institute of Technology and Management, Baroda University) – its brand identity
4. Top-level domain (TLD)	`.ac.in`	Compound TLD: **.ac** = academic institutions; **.in** = India's country-code TLD (ccTLD), managed by **NIXI**

6. URI vs URL

- **URI (Uniform Resource Identifier)** – umbrella term; **uniquely identifies** a resource, not necessarily how to reach it.
  - **URL** – identifies by **location**.
  - **URN** – identifies by unique **name**, regardless of location, e.g. `urn:isbn:0451450523`.
- **URL (Uniform Resource Locator)** – a type of URI that **identifies and locates** a resource; **always includes a protocol** (`http://`, `ftp://`).

	URI	URL
Purpose	Uniquely identify a resource	Identify **and** locate it
Protocol included?	Not always	Always
Relationship	Parent category	Sub-category of URI
Example	`urn:isbn:0451450523` (a specific book)	`https://www.google.com`

Analogy: URI = a person's identity ("John Doe"); URL = their home address ("123 Main Street, Vadodara, Gujarat").

7. Domain Names

- Every internet-connected device has a public **IP address** (IPv4 e.g. `192.0.2.172`; IPv6 e.g. `2001:db8:8b73::8a2e:370:1337`).

- IPs are hard to memorise and change over time, so we use human-readable **domain names** to represent server IPs.
- A domain has several **labels separated by dots**, and is parsed **right to left**.
- Example: `developer.mozilla.org` → TLD `org`, label 1 `mozilla`, label 2 `developer`.

TLD (Top-Level Domain)

- Tells users the general purpose of the service.
- **Generic TLDs** (`.com`, `.org`, `.net`) have no strict criteria.
- Can contain special as well as Latin characters; **max length 63 characters** (most are 2–3).
- Official list maintained by **ICANN**.
- **Restricted TLDs**
  - **Country TLDs** (`.in`, `.fr`, `.us`) may require service in a given language or hosting in that country.
  - **`.gov`** – government departments and official authorities only.
  - **`.edu`** / **`.ac`** – accredited educational and academic institutions only.

Labels

- Components that follow the TLD (reading right to left) in the domain name.
- **Length:** 1–63 characters.
- **Case-insensitive.**
- **Allowed characters:** letters A–Z, digits 0–9, hyphen (-).
- A hyphen **cannot be the first or last** character.
- Valid examples: `a`, `97`, `hello-strange-person-16-how-are-you`.

SLD and subdomains

- The label directly before the TLD is the **Second-Level Domain (SLD)**.
- Domains can have many labels, e.g. `informatics.ed.ac.uk`.
- For a domain you control (e.g. `mozilla.org`) you can create **subdomains** with unique content: `developer.mozilla.org`, `support.mozilla.org`, `bugzilla.mozilla.org`.

Buying a domain name

- You **cannot buy a domain permanently** – you pay for the **right to use** it for one or more years.
- You have priority to **renew** over other applicants, but never truly own it. This lets unused names become available again.
- **Registrars** use **registries** to track technical and administrative information linking you to your domain.
- Exception: some TLDs are managed directly by one company (e.g. every `.fire` domain is managed by Amazon, per the slide).

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8. HTTP Protocol & Header Anatomy

HTTP request (client → server)

1. **Request line:** Method (GET, POST, PUT, DELETE) + Request-URI (e.g. `/api/data`) + Protocol version (`HTTP/1.1`).
2. **Headers** (key–value metadata from the client):
   - `Host: www.itm.ac.in` – **mandatory in HTTP/1.1**
   - `User-Agent: Mozilla/5.0…` – browser identity
   - `Accept: text/html, application/json`
3. **Body:** payload for POST/PUT (form data/JSON).

HTTP response (server → client)

1. **Status line:** Version + Status code + Reason phrase (e.g. `HTTP/1.1 200 OK`, `404 Not Found`).
2. **Headers** (server metadata):
   - `Content-Type: text/html; charset=utf-8`
   - `Set-Cookie: session_id=xyz123`
   - `Cache-Control: max-age=3600`
3. **Body:** HTML document, CSS file, or API JSON payload.

9. HTTP Request–Response Lifecycle

1. **DNS resolution** – browser converts the domain (e.g. `example.com`) to an IP via: local DNS cache → recursive resolver → root / TLD servers → authoritative DNS.
2. **TCP 3-way handshake** – client sends **SYN**, server replies **SYN-ACK**, client sends **ACK**; a reliable socket connection is established on **port 80 (HTTP)** or **443 (HTTPS/SSL)**.
3. **HTTP transmission** – client sends the HTTP request; server processes routing logic and database queries, then returns the HTTP response.
4. **Connection teardown / keep-alive** – the socket closes, or stays open for further requests using **persistent connections**.

10. Web Browser vs Web Server

Web browser (user-agent engine)

Objective: fetch remote web assets, parse code, build **DOM/CSSOM trees**, render layout, execute client scripts.

- **Rendering engine:** Blink (Chrome), Gecko (Firefox), WebKit (Safari).
- **JavaScript engine:** V8 (Chrome), SpiderMonkey (Firefox).
- **Network & storage layer:** HTTP cache, LocalStorage, cookies, TLS stack.

Web server (server daemon)

Objective: listen on network sockets (`0.0.0.0:80/443`), accept HTTP requests, check access controls, serve static files or route to application runtimes.

- **Process/thread per connection:** Apache HTTP Server (MPM Prefork/Worker) – high memory use under heavy concurrent load.
- **Asynchronous, event-driven:** Nginx / Node.js – non-blocking I/O loop handling 10k+ concurrent connections efficiently.

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11. IP Addresses

Definition and working

- An **IP address** is a unique numeric label given to every device on the internet or a local network, enabling communication and location identification.

- It acts as a **locator** that helps a device find and retrieve data on the internet.
- Addresses are allocated by **IANA** (Internet Assigned Numbers Authority); the IPv4 range is `0.0.0.0` to `255.255.255.255`.
- Any connected hardware (laptops, IoT devices, smart appliances) needs an IP to route data between networks.
- Two main versions: **IPv4** and **IPv6**. Four types: **public, private, static, dynamic**.

Four types of IP addresses

Type	Description
Public	External-facing address assigned to your **router by the ISP**; all local devices use it to talk to the wider internet. Needed for port configuration, gaming, hosting email/web servers, streaming, remote connections
Private	Internal address assigned by the **router** to devices on the local network. Reused across networks, which preserves IPv4 space. In IPv6 the private equivalent is **ULA (Unique Local Addressing)**
Static	Manually configured, **permanent** address that does not change automatically. An ISP may also assign one to a user account. Used for servers/devices that need a consistent address
Dynamic	**Temporary** address assigned automatically from a pool of unassigned addresses; changes over sessions. Standard for most consumer devices

Dynamic IP and DHCP

- **DHCP (Dynamic Host Configuration Protocol)** distributes dynamic IPs. The DHCP server can be the router in a home or organisation.
- Each time a user connects, a fresh IP is assigned from the pool of currently unassigned addresses (**address cycling**).
- The IP changes on router reboot or **lease expiry**.

Static vs dynamic (server perspective)

- **Static IP:** constant; essential for production web servers, database clusters, authoritative DNS servers; more expensive.
- **Dynamic IP:** leased by ISP via DHCP; changes periodically.

Dynamic DNS (DDNS) – very important

- **Problem:** a standard DNS **A-record** maps a name to a fixed IP. If a home/small-business server's dynamic IP changes, the domain lookup **fails**.
- **Solution – DDNS:** a lightweight **daemon** runs on the server. Whenever the **WAN IP changes**, it sends an **authenticated API request** to update the DNS A-record within seconds.

12. IPv4

- Fourth version of the Internet Protocol; first deployed on the **SATNET** (Atlantic Packet Satellite Network) in the early 1980s.
- **32-bit** address written as four decimal numbers (0–255), each an 8-bit binary number, separated by dots, e.g. `10.0.0.1`.
- About **4.3 billion** (2^{32}) unique addresses – now nearly exhausted.
- Still carries most internet traffic today.

Reserved IPv4 addresses

Address	Name	Purpose
`0.0.0.0`	Default network	Non-routable meta-address meaning invalid / unknown / non-applicable target
`127.0.0.1`	Loopback	A computer addresses itself / tests local services, independent of its assigned IP
`169.254.0.1–254`	Link-local	Auto-assigned when a client **cannot get an IP from a DHCP server**
`255.255.255.255`	Broadcast	Sends a message to **every machine** on the local network segment

Address classes / subnet ranges (as given on the slides)

Class	Range	Use
A	`10.0.0.0` – `10.255.255.255`	Common private subnet
B	`172.16.0.0` – `172.31.255.255`	Common private subnet
C	`192.168.0.0` – `192.168.255.255`	Common private subnet
D	`224.0.0.0` – `239.255.255.255`	Multicast
E	`240.0.0.0` – `254.255.255.254`	Experimental

- Classes A, B, C are the most common for creating subnets. Class D follows IETF multicast rules. Releasing Class E for public use was debated before IPv6.

Subnets

- A **subnet** is a smaller network connected to a larger one through a router. Devices in a subnet talk directly without sending data through the wider internet.
- IANA reserves address blocks for companies, governments and ISPs; the ISP assigns you an address from these blocks.
- Home routers: the ISP gives **one IP to the router**, which creates a **private subnet** for all connected devices.
- Each IP in a subnet has a **network ID** and a **node (host) ID**, separated using a **subnet mask**.
- In every subnet, the **first address identifies the subnet** itself and the **last address is the broadcast address**.

13. IPv6

- Introduced because IPv4 could not handle the explosion in the number of devices.
- **128-bit** addresses in **hexadecimal**, giving about **340 undecillion** addresses (practically unlimited).
- Structure: **eight groups of four hexadecimal digits** (0–9, A–F), separated by colons, e.g. `2001:db8::1`.

Drawbacks of IPv4

1. **Scalability & capacity** – limited address space; broadcast traffic adds network load and hurts performance.
2. **Routing & efficiency** – complex manual/DHCP configuration causes errors; variable-length headers add processing overhead at each hop; fragmentation by routers slows throughput and risks packet loss.
3. **Security & performance** – no built-in security (IPsec is an add-on); limited QoS support hurts real-time apps (HD video, voice).

Benefits of IPv6

- **Larger address space** (128-bit).
- **Improved security** – IPsec (authentication and encryption) built into the design.
- **Simpler header** – fixed size, faster routing.
- **Better QoS** support for streaming.
- **Mobile-friendly** – seamless, secure support for mobile connections.

Transition strategies

- **Dual stacking** – devices run both IPv4 and IPv6 stacks at the same time.
- **Tunneling** – IPv6 packets are encapsulated inside IPv4 so they can cross non-upgraded networks.
- **NAT (Network Address Translation)** – translates address headers so different IP versions can communicate.

IPv4 vs IPv6 comparison

Feature	IPv4	IPv6
Address size	32-bit	128-bit
Notation	Decimal, dot-separated	Hexadecimal, colon-separated
Address space	~4.3 billion	Extremely large
Configuration	Manual and DHCP	SLAAC, DHCPv6, manual

Feature	IPv4	IPv6
Connectivity (NAT)	Often affected by NAT	End-to-end connectivity restored without NAT
IPsec	Optional	Built into the protocol
Fragmentation	By sender **and** routers	Only by the sender
Packet flow ID	Not supported	**Flow Label** field
Header checksum	Included	Not included
Transmission	Broadcast	Multicast and anycast (no broadcast)
Header size	Variable (20–60 bytes)	Fixed 40 bytes
Address classes	A, B, C, D, E	None
Subnetting	VLSM	Prefix-based
Example	`66.94.29.13`	`2001:db8::1`

14. IP Address vs MAC Address

	IP address	MAC address
Identifies	A device's **connection to a network**	The **physical device** (network card/router)
Assigned by	ISP	Hardware manufacturer
Size / layer	4-byte (IPv4) or 16-byte (IPv6); **Network layer**	6-byte hexadecimal; **Data Link layer**
Nature	Usually **dynamic**	**Static**, bound to the hardware
Visibility	Easily found by third parties on the internet	Private; visible only to others on your **LAN**

15. Finding Your IP Address

- **Windows:** open Command Prompt (type `cmd`) → run `ipconfig`.
- **Mac:** Apple menu → System Preferences/Settings → **Network**.
- **Mobile:** Settings → Wi-Fi/Network → select your network → IP listed under Advanced.

16. IP Security Threats and Protection

Threats (online stalking and social engineering are the two leading ones)

- **Location tracking** – narrowing down your physical area from the IP.
- **Targeted DDoS attacks** on your home network.
- **Illegal downloads** using your IP, falsely implicating you.

5 ways to protect your IP

1. **VPN** – encrypts traffic and masks location.
2. **Proxy server** – acts as an intermediary for web requests.
3. **Dynamic IP** – ISP-assigned addresses that change periodically.
4. **NAT firewall** – hides your private IP from external networks.
5. **Modem reset** – may give a new IP when the connection is re-established.

17. IP Address FAQs

- **Use of an IP:** identifies every device so computers, phones and IoT devices can communicate on the internet and LANs.
- **If someone has your IP:** hacking risk is low, but your location can be narrowed down; attackers may identify your ISP for phishing; services like Netflix/Hulu use IPs for regional content restrictions.
- **Two types of IPs:** public (external, usually the router) and private (laptops, printers, smart TVs).
- **192.168.x.x:** start of the private range up to `192.168.255.255`; `192.168.0.0` itself is not usually assigned to a device.
- **Loopback `127.0.0.1`:** lets a computer refer to itself regardless of any assigned network IP.

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18. Evolution to Web 2.0 Technologies

1. **AJAX & async processing** – `XMLHttpRequest` and the modern `fetch()` API let web apps request data in the background **without full page reloads**, enabling rich **Single Page Applications (SPAs)**.
2. **User-generated content** – shift from read-only static sites to participatory platforms (social networks, wikis, blogs) where users publish, comment and collaborate.
3. **RESTful APIs & services** – lightweight **JSON** exchanges over HTTP REST APIs, replacing heavy **SOAP/XML**; enables third-party **mashups**.
4. **Software as a Service (SaaS)** – apps moved from installed desktop binaries to cloud-hosted browser apps (Google Docs, Figma, Office 365).

19. Linux Web Server Setup (Apache & Nginx)

Step	Task	Detail
1	Package installation	`sudo apt update && sudo apt install apache2 nginx` on Debian/Ubuntu
2	Directory & permissions	Web document root at `/var/www/html`; assign ownership to the server user with `chown`
3	Virtual host configuration	Create domain config files in `sites-available` to **host multiple domains on a single IP**
4	Firewall & SSL	Open HTTP/HTTPS ports with `ufw`; issue SSL certificates with **Let's Encrypt Certbot**

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20. Web Design Fundamentals

> "Design is not just what it looks like and feels like. Design is how it works." – Steve Jobs

Core concepts of UI/UX

- **User-centricity** – prioritise end-user needs; know the target audience; build intuitive flows that reduce friction.

- **Visual hierarchy** – use size, colour, spacing, weight and contrast to guide the eye and show the most important information first.
- **Accessibility (a11y)** – usable by everyone, including people with visual, motor or cognitive disabilities, following **WCAG** standards.

Responsive Web Design (RWD)

- A responsive site **fluidly adjusts layout, typography and images** to any screen, from large desktop to small smartphone.
- Achieved with **fluid grids**, **CSS media queries**, **Flexbox** and **CSS Grid**.

Visual hierarchy and the F-pattern

- Users **scan** pages rather than read word-for-word.
- Eye-tracking shows an **F-shaped scan pattern**, focusing on the **top and left**.
- Put logos, primary navigation and key calls-to-action along these paths.
- Use a **Z-pattern** for visual landing pages.
- A clear hierarchy prevents users from getting lost in a wall of text.

Navigation and the 3-click rule

- Clear site maps, breadcrumbs and persistent header menus.
- Users should reach any content within about **3 intuitive clicks** from the home page.

Accessibility (WCAG 2.1)

- Minimum **4.5:1 text colour contrast ratio**.
- Provide **alt attributes** for images.
- Ensure full **keyboard navigation**.

21. Web Design Challenges

1. **Responsive design** – fluid layouts across smartphones (~360px), tablets (~768px) and 4K displays using media queries, Flexbox, Grid.
2. **Cross-browser compatibility** – differences between engines (Chromium vs Gecko vs WebKit); solved with CSS resets, **Autoprefixer** (vendor prefixes) and feature detection via **Modernizr**.

3. **Performance & speed** – optimise **Core Web Vitals (LCP, CLS, FID)**; use **WebP** compression, CSS/JS **minification**, **CDNs** and async loading.
4. **Security** – prevent **XSS (Cross-Site Scripting)**, mitigate **clickjacking**, enforce **Content Security Policy (CSP)** headers.

22. Common Design Issues

- **Slow load times** – conversion rates fall as load time rises (slide data): **1 s → 3.2%**, **2 s → 1.7%**, **3 s → 1.1%**, **4 s → 0.9%**.
- **Mobile unresponsiveness** – forcing pinch-and-zoom creates friction, high abandonment, poor brand perception and **SEO penalties**.
- **Cluttered layouts** – too much text/images/features causes cognitive overload; **whitespace is crucial**.
- **Poor accessibility** – low contrast and tiny fonts make sites unusable for many people.
- **Technical glitches** – broken links, outdated code, server errors damage trust and search indexing.

23. Best Practices Checklist

- **Mobile-first approach** – design for mobile devices first.
- **Prioritise performance** – optimise images, minimise code, use caching.
- **Clear visual hierarchy** – typography, whitespace and colour guide the eye.
- **Ensure accessibility** – alt text, high contrast, keyboard-friendly navigation.

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24. Exam Review Questions (from the slides)

1. **Differentiate between the Internet and WWW with suitable examples.** (4 marks) – see Section 4.
2. **Explain the HTTP request–response header structure in detail.** (6 marks) – see Sections 8 and 9.
3. **Describe the mechanism of Dynamic IP and how Dynamic DNS resolves IP shifts.** (4 marks) – see Section 11.
4. **What are the key features of Web 2.0? Explain AJAX.** (5 marks) – see Sections 2 and 18.

Link to the lab: the server setup and HTTP concepts form the baseline for deploying HTML/CSS projects in **Lab Experiments P1–P5**.

Next lecture (Unit 2): HTML5 fundamentals – document type, formatting tags, forms, HTML5 media elements and semantic tags.

Quick Revision Sheet

- Internet = network (TCP/IP); WWW = service on top (HTTP + URIs + HTML).
- URL ⊂ URI. URN names, URL locates.
- Domain is read **right to left**: TLD → SLD → subdomain.
- HTTP request = request line + headers + body; response = status line + headers + body.
- Default ports: **80** (HTTP), **443** (HTTPS).
- Lifecycle: DNS → TCP handshake (SYN, SYN-ACK, ACK) → HTTP → keep-alive/close.
- Apache = process/thread per connection; Nginx/Node.js = async event-driven.
- IPv4 = 32-bit; IPv6 = 128-bit. MAC = 6-byte physical address (Data Link layer).
- Dynamic IP changes → **DDNS** updates the DNS A-record automatically.
- Web 2.0 = AJAX + UGC + REST/JSON + SaaS.
- Design: mobile-first, fast, clear hierarchy, accessible (WCAG 4.5:1 contrast).

Notes on Slide Errors (worth knowing for exams)

- The slide lists Class C as `192.186.0.0`; the correct private range starts at **`192.168.0.0`**.
- The "Class A/B/C" ranges on the slides are really the **private address ranges** (10.x, 172.16–31.x, 192.168.x). In classful addressing, Class A actually spans `1.0.0.0–126.255.255.255`, Class B `128.0.0.0–191.255.255.255`, Class C `192.0.0.0–223.255.255.255`.
- Class E is normally given as `240.0.0.0–255.255.255.255` (the last address, `255.255.255.255`, is the limited broadcast address).
- The IPv4 vs IPv6 graphic shows "deployed 1981 / 1998" while another slide says IPv4 first ran on SATNET in 1982; use whichever your faculty prefers (the commonly cited year is 1981–83).
- IPv6 is not literally "unlimited" – it is just so large (2^{128}) that exhaustion is not a practical concern.